Jonathan David Juarez

• Arcata, CA • JDJ58@humboldt.edu • (707)362-3778

Education

Bachelor of Science: Data Science | Economics (in progress) GPA: 3.59
California State Polytechnic University, Humboldt Expected May 2027

Relevant Coursework:

- **Data 322: Machine Learning**
- **Math 313: Ordinary Differential Equations**
- **Math 344: Linear Algebra**
- **Stat 333: Linear Regression**
- **CS 325: Database Design**
- **Econ 306: Econ. of The Developing World**
- **Econ 309: Econ. of a Sustainable Society**
- **Econ 310: Intermediate Micro Strategy**
- **Econ 311: Intermediate Microeconomics**
- **Econ 435: Financial Inst. & Cap. Markets**

Research Experience

Researcher Lawrence, KS | May 2026-Present
University of Kansas, The Kansas Data Science Consortium Department of Economics

- Engineered and harmonized a balanced panel dataset of 284,280 observations (2003–2022) in Stata, cleaning geographic boundary anomalies across U.S. counties and mapping spatial treatment histories using FEMA disaster declarations.
- Implemented advanced event-study designs, deploying the Sun-Abraham estimator to correct for parallel trend instabilities and weighting biases present in canonical Two-Way Fixed Effects (TWFE) models.
- Uncovered distinct funding lags and capacity shifts across state and federal channels, while laying the econometric groundwork for structural mediation models to isolate how institutional resilience affects community migration outcomes.

Researcher Arcata, CA | August 2025-Present
California State Polytechnic University, Humboldt Department of Mathematics and Data Science

- Investigated wildfire-driven forest loss in Humboldt County's redwood ecosystems in Six Rivers National Forest, with a focus on lightning-induced fires and climate stressors such as drought and vegetation change.
- Analyzed geospatial datasets including PDSI, NDVI, Wildfire perimeters, severity, and forest loss (Hansen et al., Cal Fire) to quantify forest degradation and identify spatial patterns of wildfire impact.
- Established a spatial and temporal foundation for predictive modeling by identifying a post-2000 increase in lightning-caused fires, supporting future machine learning-based forest loss prediction and data-driven forest management.

Researcher Stanford, CA | June 2025-August 2025
Stanford University Doerr School of Sustainability SURGE

- Developed and presented research on atmospheric data and sustainability by investigating how accessible, accurate, and globally available datasets can reveal what lies beneath our skies and inform climate-related decision-making.
- Led collaborative undergraduate research by designing approaches, asking critical research questions, and documenting methods in a research journal, while also mentoring and motivating student researchers.
- Expanded skills at the intersection of data science, remote sensing, and climate action, gaining hands-on experience with data-driven methods while building connections with Stanford faculty, PhD students, alumni, and peers.

Research Experience Continued

Researcher Arcata, CA | August 2024-June 2025
UC-CSU Environmental and Climate Change Leadership Program

- Conducted in-depth research on climate change literacy trends, evaluating current curriculum materials and identifying gaps in environmental education.
- Analyzed data from surveys and interviews to assess the effectiveness of climate change awareness programs across diverse populations.
- Worked closely with faculty members to provide evidence-based recommendations for enhancing environmental sustainability and literacy within university programs.

Researcher Arcata, CA | January-May 2025
California Polytechnic University, Humboldt Department of Mathematics and Data Science

- Conducted a longitudinal data analysis (2000–2024) integrating public employment sector wages and rental market data to evaluate housing affordability trends using Python, Pandas, and Plotly Express.
- Developed a multi-panel interactive visualization tool enabling users to filter by industry sector, title, and down payment level to simulate housing affordability scenarios.
- Collaborated with peers to present findings at California State Polytechnic University, Humboldt's Day of Data Science, translating complex statistical relationships into accessible, real-time visual insights for academic and public audiences.

Researcher Arcata, CA | October-December 2024
Louis Stokes Alliance for Minority Participation, California State Polytechnic University, Humboldt

- Utilized Google Earth Engine and JavaScript to analyze geospatial data for wildfire risk assessment.
- Conducted vegetation health analysis using NDVI and other indices to identify areas with high fire susceptibility.
- Developed interactive GIS maps and models to visualize and communicate risk factors effectively.

Work Experience

Marine Machine Learning & Data Science Intern Blue Lake, CA | August 2026-Present
Marine Applied Research and Exploration

- Engineered media processing pipelines utilizing FFmpeg and custom Python scripts to batch-convert legacy DVD-RW media into standardized MP4 formats, automated thumbnail and poster generation, and mapped network drives for seamless deployment to a Jellyfin media server.
- Restructured and audited two legacy database systems, executing data standardization protocols to rectify historical derivation errors, establishing strict user permissions to ensure future data integrity, and authoring comprehensive README documentation for all core variables.
- Architected a foundational training dataset deployed in machine learning classification models for marine fish and invertebrates, enabling a non-invasive methodology to track and identify wildlife populations within underwater transects.

Assistant Editor Arcata, CA | February 2025-Present
Humboldt Economic Index

- Contributed to the monthly production of the Humboldt Economic Index by collecting, cleaning, and analyzing regional economic indicators such as employment, home sales, lumber prices, and consumer confidence.
- Authored and edited data-driven reports and economic summaries, ensuring clarity, accuracy, and alignment with trends relevant to Humboldt County's local economy.
- Collaborated with faculty advisors and student researchers to synthesize quantitative findings into publicly accessible insights, supporting regional economic literacy and decision-making.

Co-Director Arcata, CA | August 2024-August 2026
Reusable Office Supply Exchange (ROSE)

- Oversaw teams within Waste Reduction Resource Awareness Program (WRRAP), such as Zero Waste, Compost, and Bicycle Learning Center; recruited, trained, and managed volunteers to implement initiatives, events, and educational campaigns on waste reduction and sustainability.

Work Experience Continued

- Planned and executed projects aligned with WRRAP goals; developed long-term strategies for growth; managed budgets, secured funding, and prepared comprehensive reports for the institution.
- Organized campus-wide sustainability events such as workshops, waste audits, and Zero Waste Conference that represented WRRAP, fostering relationships with campus and local sustainability-focused organizations to promote its mission.

Dealer Support Carson, CA | July 2021-August 2024
Toyota Motors North America

- Analyzed sales data to generate daily, weekly, and monthly reports for upper management, supporting decision-making and forecasting.
- Collaborated with vendors and project managers to manage inventory and ensure compliance with Toyota manufacturing and dealership standards.
- Streamlined processes, improved standard operating procedures, managed accounts and billing, and conducted troubleshooting for WMS and ordering portals in controlled environments.

Language Fluency

- **English,** Native Speaker
- **Spanish,** Fluent
- **French**, Conversational

Certifications

Business Analytics with Excel March 2024
Microsoft
Career Essentials in Business Analysis March 2024
Microsoft
IBM Data Analyst October 2022
IBM
Google Information Technology Automation April 2022
Google
Information Technology Support November 2021
Google

Skills

Python
Java
SQL
Google Earth Engine
ArcGIS|QGIS

Professional Affiliations

Ecological Forecasting Initiative (EFI) May 2026 - Present
Mathematical Association of America (MAA) September 2025 - Present
Society for Advancement of Chicanos and Native Americans in Science March 2025 - Present
Society of Hispanic Professional Engineers (SHPE) April 2024 - Present

Honors

Dean's Honor List, California State Polytechnic University, Humboldt December 2026
Presidential Scholar, California State Polytechnic University, Humboldt December 2025
McNair Scholars Program, California State Polytechnic University, Humboldt September 2024 - Present
Dean's Honor List, California State Polytechnic University, Humboldt December 2024

Honors Continued

Dean's Honor List, Los Angeles Harbor College, Wilmington, CA December 2023
Dean's Honor List, Los Angeles Harbor College, Wilmington, CA May 2023

Invited Talks and Presentations

Presenter, *Ecological Forecasting Initiative 2026 Annual Conference* – Arcata, CA August 2026
Presented a geospatial assessment of wildfire-driven forest loss using CAL FIRE ignition records (1950–2025), Hansen Global Forest Change data, MODIS burn-severity layers, NDVI, and PDSI. Identified strong spatial overlap between lightning-caused ignitions and high-loss forest clusters, with a marked rise in lightning-driven fires after 2000. Demonstrated how climate-linked ignition patterns and vegetation structure interact to shape long-term forest vulnerability.

Presenter, *IdeaFest 2026* – Arcata, CA May 2026
Developed and presented a continuous-time model of smartphone battery discharge incorporating activity-dependent current demand and temperature-driven degradation. Simulated battery performance across common usage scenarios and varying states of health, demonstrating that high-energy activities and elevated temperatures significantly accelerate depletion. Validated model predictions against benchmark datasets, establishing a framework for estimating battery life and evaluating user-behavior impacts on device longevity.

Presenter, *IdeaFest 2026* – Arcata, CA May 2026
Conducted a multi-source analysis of post-graduation employment outcomes across academic majors, integrating salary distributions, geographic clustering, skill requirements, and early- vs. mid-career trajectories. Led data curation, cleaning, and exploratory modeling to identify structural patterns in labor-market opportunities and discipline-specific career mobility. Presented findings highlighting disparities across fields and the predictive value of skill alignment in long-term earnings.

Presenter, *IdeaFest 2026* – Arcata, CA May 2026
Presented a geospatial assessment of wildfire-driven forest loss using CAL FIRE ignition records (1950–2025), Hansen Global Forest Change data, MODIS burn-severity layers, NDVI, and PDSI. Identified strong spatial overlap between lightning-caused ignitions and high-loss forest clusters, with a marked rise in lightning-driven fires after 2000. Demonstrated how climate-linked ignition patterns and vegetation structure interact to shape long-term forest vulnerability.

Speaker, CSU Research Competition 2026 - San Jose, CA April 2026
Presented a geospatial assessment of wildfire-driven forest loss using CAL FIRE ignition records (1950–2025), Hansen Global Forest Change data, MODIS burn-severity layers, NDVI, and PDSI. Identified strong spatial overlap between lightning-caused ignitions and high-loss forest clusters, with a marked rise in lightning-driven fires after 2000. Demonstrated how climate-linked ignition patterns and vegetation structure interact to shape long-term forest vulnerability.

Presenter, McNair Research Conference 2026 - Baltimore, MD March 2026
Presented a geospatial assessment of wildfire-driven forest loss using CAL FIRE ignition records (1950–2025), Hansen Global Forest Change data, MODIS burn-severity layers, NDVI, and PDSI. Identified strong spatial overlap between lightning-caused ignitions and high-loss forest clusters, with a marked rise in lightning-driven fires after 2000. Demonstrated how climate-linked ignition patterns and vegetation structure interact to shape long-term forest vulnerability.

Presenter, *SACNAS NDiSTEM 2025 Annual Conference* - Columbus, OH October 2025
Presented preliminary findings from a geospatial analysis of wildfire-driven forest loss in Humboldt County's redwood-dominated ecosystems. Using Hansen Global Forest Change data, MODIS burn-severity layers, and CAL FIRE ignition perimeters, the study identified pronounced spatial overlap between high-severity wildfire zones and subsequent forest-loss clusters. Temporal analysis showed a notable post-2000 increase in lightning-caused ignitions, marking a significant shift in regional wildfire dynamics. These results establish the empirical foundation for the project's next phase: the development of a machine-learning model that integrates NDVI, PDSI, pre-fire

Invited Talks and Presentations Continued

drought stress, vegetation structure, and binary wildfire-occurrence data to predict future forest loss and highlight emerging high-risk areas for climate-adaptive forest management.

Guest Speaker, DSEPodcast October 2025
Discussed the application of data science in real-world challenges, including *Housing and Wages in Humboldt County* at California State Polytechnic University, Humboldt, and *Near-Real-Time Monitoring of Gridded Livestock Methane Emissions Using Multi-Source Data Integration* at Stanford University. Addressed the importance of data availability in research and the role of machine learning in filling gaps, as well as how sustainability issues intersect with social justice, such as urban greening and child labor in supply chains.

Keynote Speaker, Tech Partnership Summit (TPS) October 2025
I delivered a keynote on "From Classroom To Climate Impact" to an audience of professional background attendees. The talk focused on how my academic and research experiences have shaped my understanding of data science, with a particular emphasis on the intersection of coursework, hands-on research, and industry-standard tools.

Speaker, *University of New Mexico McNair Scholars Research Conference,* NM September 2025
Delivered a presentation on *Near-Real-Time Monitoring of Gridded Livestock Methane Emissions Using Multi-Source Data Integration* at Stanford University, in collaboration with Kim Nguyen, Dr. Xinyu Duo, and Dr. Rob Jackson. This study focused on developing an improved bottom-up inventory model for livestock methane emissions, integrating multi-source activity data from FAOSTAT, USDA, and other databases to generate near-real-time, gridded methane estimates. The research aimed to address data gaps, improve temporal resolution, and provide more precise methane tracking, enabling better climate reporting and policy applications.

Presenter, *Day of Data Science, California State Polytechnic University, Humboldt,* CA May 2025
Presented research on *Housing and Wages in Humboldt County: A Quarter-Century Analysis* in collaboration with Devon Iris Walker. This project analyzed historical housing trends in Humboldt County, focusing on the disparity between wage growth and rising rental prices. By leveraging data science techniques and datasets from sources such as the Bureau of Labor Statistics, Freddie Mac, and Humboldt State University's Economic Index, we examined the impact of the Great Financial Crisis and COVID-19 on housing affordability. Key research questions addressed included the rent-to-income ratio across industries, wage trends relative to rental prices, and the effects of economic disruptions on housing affordability.

Presenter, *2025 Gabriel E. Gallardo Research, Student Leadership & Advocacy Symposium,* WA April 2025
Research Title: *Remote Sensing for Forest Health and Wildfire Risk Mitigation*
Presented research on Remote Sensing for Forest Health and Wildfire Risk Mitigation, exploring how satellite imagery and climate data (MODIS, Landsat, GRIDMET) can be used to assess forest health and predict wildfire risks. The study found increased vulnerability to drought stress in northern Humboldt County, challenging assumptions about regional moisture levels and highlighting the need for tailored forest management strategies to address wildfire risks.

Awards and Scholarships

Travis Jepsen Memorial Mathematics Scholarship May 2026
College of Natural Resources & Sciences (CNRS), California State Polytechnic University, Humboldt

- Awarded $3,000 scholarship recognizing academic excellence in Mathematics with a GPA above 3.0.

Excellence in Academic Discipline for the College of Natural Resources & Sciences April 2026
California State Polytechnic University, Humboldt

- Recognized for outstanding academic achievement in Data Science and Economics within the College of Natural Resources & Sciences.
- Demonstrated excellence in interdisciplinary coursework, analytical problem-solving, and academic performance.

Awards and Scholarships Continued

Outstanding Student Researcher April 2026
California State Polytechnic University, Humboldt

- Awarded for exceptional undergraduate research contributions in Data Science and Economics.
- Recognized for research excellence, analytical rigor, and academic engagement across interdisciplinary projects.

Atkinson Family Scholarship April 2026
College of Professional Studies (CPS), California State Polytechnic University, Humboldt

- Awarded a competitive $1,000 scholarship for academic achievement and good standing as a junior/senior Economics student.
- Recognized for strong academic performance and commitment to the field of Economics.

ADPI+MENA Scholar Program, California State Polytechnic University, Humboldt February 2026 - Present

- Participating in a structured scholar program focused on academic mentoring, professional development, and career readiness.
- Receiving academic coaching, advising, and skill development in research, statistics, and career preparation.
- Developing a community impact project aligned with academic and professional goals.

STEM Star Program - California State Polytechnic University, Humboldt August 2025 - December 2025
STEM Student Advancement through Research and Support

- Competitive, merit-based academic development program for high-achieving STEM students
- Selected for placement in a faculty-mentored research cohort
- Participated in professional development workshops focused on academic and career readiness
- Engaged in a structured peer Field-Intensive Team to build academic capital and professional identity
- Culminated in a public research presentation supporting long-term STEM academic and career pathways

SACNAS Research Presentation Award- Columbus, Ohio November 2025
Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)

- Recognized for excellence in research presentation at the SACNAS National Diversity in STEM (NDISTEM) Conference.
- Presented original research titled “Remote Sensing for Forest Health and Wildfire Mitigation.”

SACNAS SACRED Mentoring Program October 2025

- SACNAS: STEM Advancement for Community, Research, Education, and Development
- Competitive six-month mentoring and professional development program emphasizing Vision, Strength, Connection, and Impact.
- Engaged in structured mentor-to-mentee activities, peer-to-peer leadership development, and community-centered STEM support.
- Participated in training focused on fostering inclusive mentorship, empowering historically excluded STEM scholars, and cultivating long-term professional growth within the SACNAS network.

SACNAS NDISTEM 2025 Travel Scholarship October 2025
Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
Columbus, Ohio

- Awarded a competitive travel scholarship covering flight and lodging to present original research at the SACNAS National Diversity in STEM (NDISTEM) Conference.
- Selected as the sole representative from my institution to attend and present at this national conference.

Michael Jerome Hughes Memorial Scholarship Fund September 2025

- Awarded a $2,000 scholarship supporting students majoring in Data Science.
- Selected based on academic excellence and potential to contribute to the field of data science; funding amounts and recipients are determined annually based on available endowment support.

Awards and Scholarships Continued

Homer Balabanis Economics Scholarship May 2025

- Awarded a $1,000 scholarship in honor of Dr. Homer Balabanis, founding economist at Humboldt.
- Recognized for outstanding academic achievement as an Economics student.

SHPE Scholarship Award September 2024

Society of Hispanic Professional Engineers (SHPE)

- Awarded a $2,000 merit-based scholarship in recognition of academic achievement and commitment to advancing Hispanic representation in STEM fields.

Leadership Positions

President, Data Science Club January 2025 - Present

California State Polytechnic University, Humboldt

- Lead organizational strategy, project coordination, and member engagement for a student organization focused on data science, analytics, and technical skill development.
- Oversee planning of collaborative projects, workshops, and academic initiatives (e.g., poster projects, technical sessions, or Ideafest preparation).
- Coordinate with faculty and the Department of Mathematics and Data Science to support student professional development.

Treasurer, **Business & Economics Club (B&E)** January 2025 - Present

California State Polytechnic University, Humboldt

- Manage budgeting, financial tracking, and allocation of club funds for events and programming.
- Maintain transparent financial records and assist in funding requests and financial planning.
- Support event logistics and interdisciplinary collaboration between economics and business students.

Competitions

Presenter, CSU Research Competition April 2026

San Jose State University

Presented research on lightning-driven wildfire risk and forest loss in Six Rivers National Forest, integrating long-term wildfire ignition records (1950–2025) with satellite-based remote sensing data. Conducted temporal trend analysis of CAL FIRE ignition data, identifying an increasing proportion of lightning-caused fires after 2000. Performed spatial overlay analysis using Hansen Global Forest Change and MODIS burned area datasets (2014–2024), revealing strong correspondence between wildfire occurrence and forest canopy loss. Incorporated environmental indicators including Palmer Drought Severity Index (PDSI) and Normalized Difference Vegetation Index (NDVI) to assess vegetation stress and fire dynamics. Established a geospatial framework for future machine learning models to predict wildfire-driven forest loss and inform climate-adaptive forest management strategies.

Participant, NorCal DataFest - American Statistical Association April 2026

California State University, Chico

Collaborated in a 48-hour team-based data analysis competition using a real-world dataset from a Kansas healthcare system. Developed a county-level *Medical Access Score* integrating healthcare infrastructure, transportation availability, financial resources, and social factors, normalized by patient population. Conducted geospatial and quantitative analysis to identify disparities in healthcare access, highlighting underserved regions such as Jackson, Pottawatomie, and Nemaha counties. Findings suggested that lower access scores may correlate with poorer patient outcomes, while also revealing that high-scoring areas can still face provider-to-patient ratio challenges.

Participant, Mathematical Contest in Modeling (MCM) 2026

California State Polytechnic University, Humboldt February 2026

Developed a physically grounded continuous-time model of smartphone lithium-ion battery dynamics to predict state of charge (SOC) and time-to-empty under realistic usage conditions. Formulated SOC as an integral of current over time and constructed a baseline linear discharge model calibrated to empirical GSMArena benchmarks, achieving exact agreement with observed runtimes. Extended the framework to incorporate state of health (SOH), modeling capacity degradation via cycle-based and nonlinear aging functions, and introduced

Competitions Continued

temperature-dependent aging using an Arrhenius-based acceleration factor. Conducted sensitivity analysis across usage modes (calls, video, web, gaming) and health states, identifying key drivers of rapid battery depletion. Evaluated model strengths and limitations, highlighting tradeoffs between physical interpretability and predictive generalization, and proposed extensions for nonlinear behavior, thermal coupling, and real-world variability.